The University of Azad Jammu & Kashmir, Muzaffarabad

Department of Software Engineering

Software Construction & Development (Lab)

Lab 02

Task Title:	**Activity & State Machine Modeling**
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Roll No:	**2024-SE-03**
Reg Number:	**2024-UMDB-004738**

Task 1: Activity Diagram with Swimlanes – Foodpanda App

Problem Statement: Draw an activity diagram visualizing the business process of the Foodpanda application (online food ordering and delivery system) using swimlane partitions to allocate responsibilities among interacting entities.

1.1 Swimlane Roles & Responsibilities

- **Customer Swimlane:** Responsible for browsing restaurant menus, adding selected items to cart, checking out, executing payment authorization, and receiving the delivered order.
- **Foodpanda App / System Swimlane:** Serves as the coordinating platform that verifies digital payments, automatically routes orders to restaurant dashboards, and dispatches the closest active rider.
- **Restaurant Swimlane:** Receives real-time order alerts, confirms item preparation, cooks food, and securely packages items ready for rider pickup.
- **Delivery Rider Swimlane:** Accepts delivery tasks on the rider mobile application, travels to the restaurant, collects the order, and transports it to the customer location.

1.2 Process Workflow Description

The ordering process initiates when the customer launches the application, browses available vendors, and places an order with payment authorization. The Foodpanda platform processes payment validation and notifies the vendor. Concurrently, upon meal packaging by the kitchen, the system assigns a delivery courier. The rider collects the order from the counter and navigates to the customer's delivery address, concluding the activity flow.

1.3 Activity Diagram

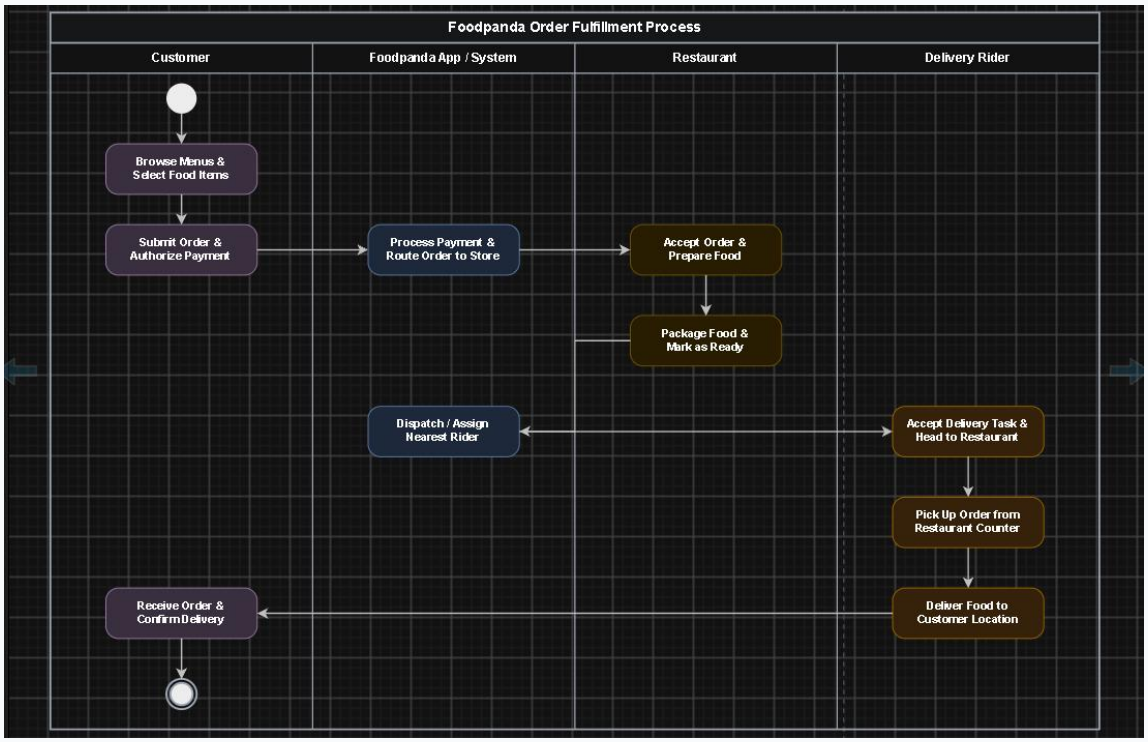


Figure: Activity Diagram with Swimlanes - Foodpanda Process

Task 2: State Machine Diagram – Toaster System

Problem Statement: Draw a State Machine diagram for a Toaster. Explicitly list internal activities (entry, do, exit) within each operating state, and categorize all triggered versus triggerless events and actions.

2.1 Operating States & Internal Activities

- **Idle State:** Represents the resting state when power is supplied but no toasting cycle has started.

• entry / resetTimer(): Resets toasting duration counter to zero.
• do / displayReadyStatus(): Keeps ready indicator illuminated.

- **Heating / Toasting State:** Active state when the carriage is held down and heating elements are energized.

• entry / engageLatch(): Solenoid secures the bread carriage down.
• entry / powerHeatingCoils(): Delivers current to internal resistive heating wires.
• do / runCountdownTimer(): Decrements duration counter towards selected browning level.
• exit / cutPowerToCoils(): Shuts off current immediately upon state departure.

- **Popping_Up State:** Transitional state where bread is ejected after heat cycle completion or user cancellation.

• entry / releaseLatch(): Solenoid releases spring-loaded mechanical carriage.
• do / raiseCarriage(): Spring moves bread carriage upward out of heating chamber.

2.2 Classification of Events & Actions

- **Triggered Events (External explicit signals):**

1. pressLever [breadPresent]: Fired externally by the user depressing the mechanical lever. Transitions system from Idle to Heating.
2. pressCancel: Fired explicitly when user presses the cancel/eject button. Forces immediate premature transition from Heating to Popping_Up.

- **Triggerless / Completion Events (Internal automatic triggers):**

1. timerExpired [Triggerless / Time Event]: Generated automatically when countdown timer reaches zero without user intervention. Transitions system from Heating to Popping_Up.
2. carriageRaisedFully [Triggerless / Completion Event]: Generated automatically by microswitch limit when carriage completes mechanical upward stroke. Automatically returns toaster from Popping_Up to Idle.

2.3 State Machine Diagram

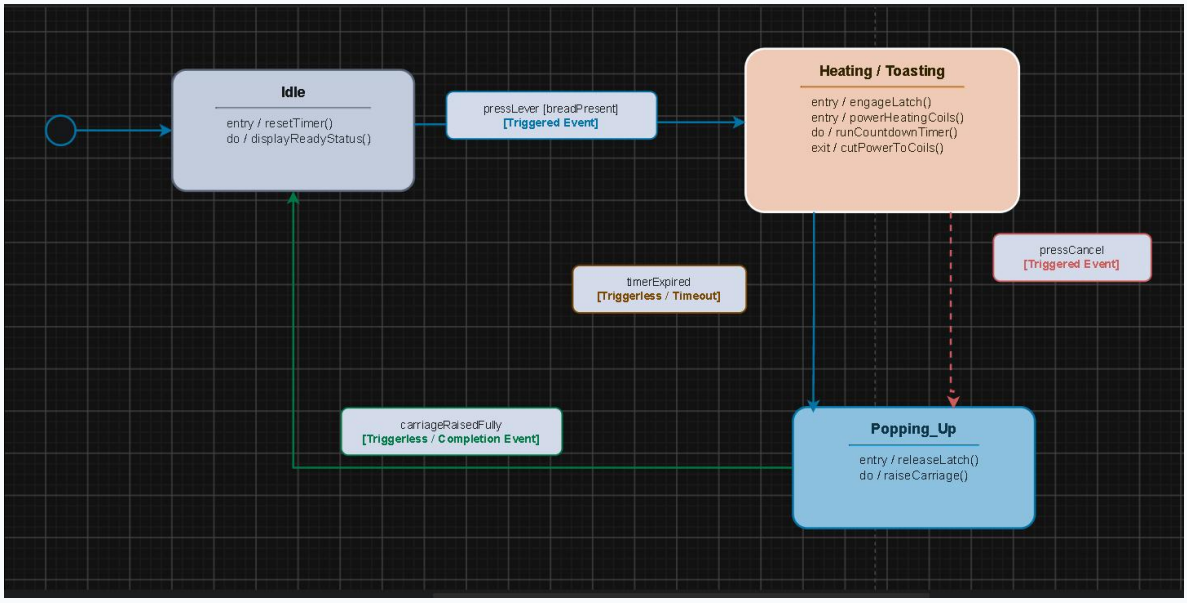


Figure: State Machine Diagram - Toaster System

Task 3: State Machine Diagram – Elevator / Lift System

Problem Statement: Draw a comprehensive State Machine diagram for an Elevator/Lift system modeling motion dynamics, door interlocks, safety sensor checks, and dispatch logic.

3.1 Elevator States & Operational Dynamics

- **Idle (Doors Closed):** Elevator rests safely at a floor level with mechanical brakes engaged and doors locked shut, awaiting call requests.
- **Moving_Up:** Brake is released and traction motor drives elevator upward toward requested destination floor.
- **Moving_Down:** Brake is released and traction motor drives elevator downward toward requested destination floor.
- **Doors_Opening:** Upon reaching target floor alignment, door motor actuator opens cabin and shaft doors.
- **Doors_Open:** Doors remain stationary in fully open position for a predetermined dwell interval (e.g., 5 seconds) while chime rings for passenger boarding.
- **Doors_Closing:** Door motor operates in reverse to shut doors. Actively checks infrared optical sensors for passenger obstruction; if an obstacle is sensed, doors immediately reverse to Doors_Open.

3.2 State Transitions & Safety Guard Conditions

- **Floor Dispatch:** requestFloor > currentFloor triggers Moving_Up; requestFloor < currentFloor triggers Moving_Down.
- **Target Arrival:** reachedTargetFloor sensor trip halts motor and triggers transition to Doors_Opening.
- **Door Safety Loop:** dwellTimerExpired triggers Doors_Closing. If obstacleDetected occurs during closing, safety guard reverses motion back to Doors_Open immediately.

3.3 State Machine Diagram

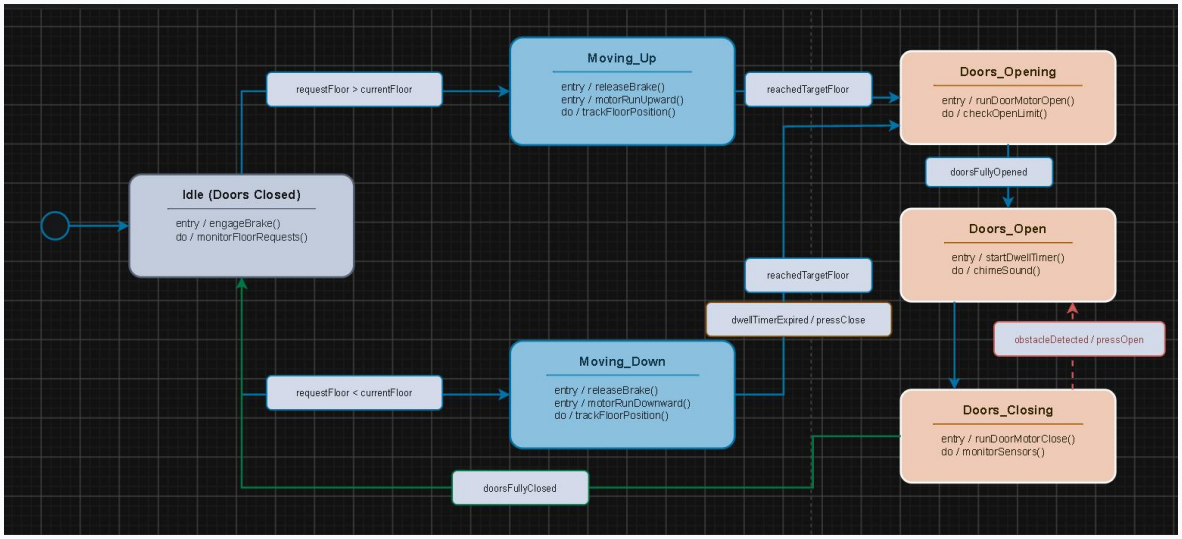


Figure: State Machine Diagram - Elevator / Lift System